the conscious part of the mind is not in high gear. A good example was Stephen Hawking's 1970 discovery, as he was getting ready for bed, that the areas of black-hole horizons must always increase (Chapter 12). Another example is a discovery by Roger Penrose that changed our understanding of what is inside a black hole.

One day in the late autumn of 1964, Penrose, by then a professor at Birkbeck College in London, was walking toward his office with a friend, Ivor Robinson. For the past year, ever since quasars were discovered and astronomers began speculating that they are powered by stellar implosion (Chapter 9), Penrose had been trying to figure out whether singularities are created by realistic, randomly deformed, imploding stars. As he walked and talked with Robinson, his subconscious was mulling over the pieces of this puzzle—pieces with which his conscious mind had struggled for many many hours.

As Penrose recalls it, "My conversation with Robinson stopped momentarily as we crossed a side road, and resumed again at the other side. Evidently, during those few moments an idea occurred to me, but then the ensuing conversation blotted it from my mind! Later in the day, after Robinson had left, I returned to my office. I remember having an odd feeling of elation that I could not account for. I began going through in my mind all the various things that had happened to me during the day, in an attempt to find what it was that had caused this elation. After eliminating numerous inadequate possibilities, I finally brought to mind the thought that I had had while crossing the street."

The thought was beautiful, unlike anything ever seen before in relativity physics. Carefully over the next few weeks Penrose manipulated it, looking at it from this direction and then from that, working through the details, making it as concrete and mathematically precise as he could. With all details in hand, he wrote a short article for publication in the journal *Physical Review Letters*, describing the issue of singularities in stellar implosion, and then proving a mathematical theorem.

Penrose's theorem said *roughly* this: Suppose that a star—any kind of star whatsoever—implodes so far that its gravity becomes strong enough to form an *apparent horizon*, that is, strong enough to pull outgoing light rays back inward (Box 12.1). After this happens, nothing can prevent the gravity from growing so strong that it creates a singularity. Consequently (since